

Polymorphism

INTRODUCTION TO OBJECT-ORIENTED PROGRAMMING IN JAVA

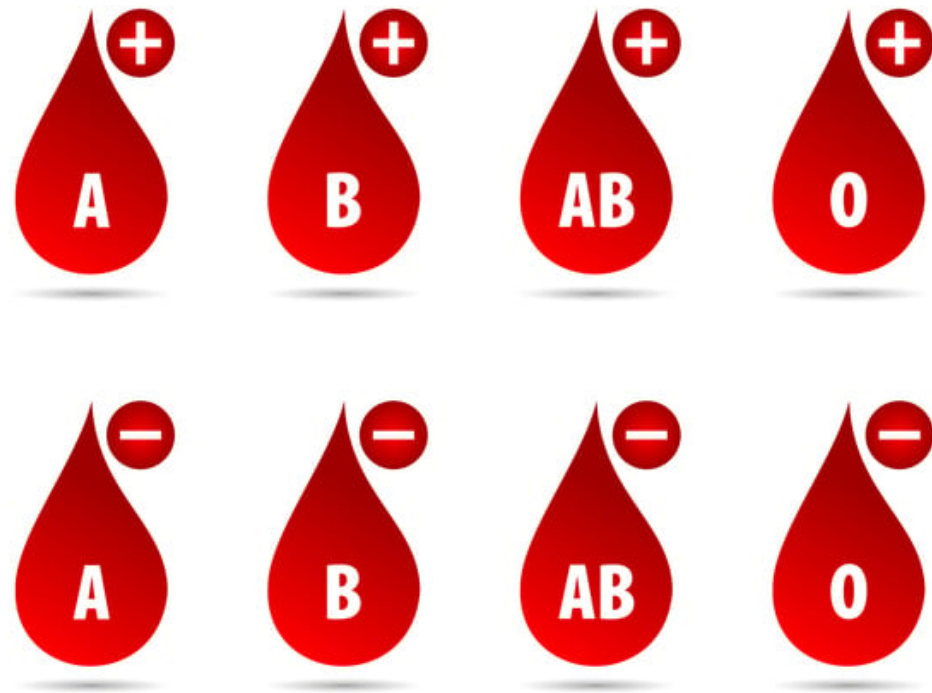


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What is Polymorphism?

- Presence of more than one form or morphs of one species
- Animal example
 - Dog
 - Cat
 - Snake
 - ...



Method overriding and method overloading

- Method overriding (runtime polymorphism)
 - @Override

```
@Override
public String toString() {
    // Print the fields info
    return "Book{title='" + ____ + "', isRented=" + ____ +
        ", author='" + ____ + "', pages=" + ____ + "}";
}
```

Method overloading

- Allow us to have multiple methods with the same name but different parameter list

```
public void greet(String animal, String sound, String habitat) { // three parameters
    System.out.println("The " + animal + " makes a " + sound +
        " sound and lives in the " + habitat + ".");
}

public void greet(String animal, String sound) { // two parameters
    greet(animal, sound, "unknown habitat");
}

public void greet() { // no parameters
    greet("Animal", "unknown sound", "unknown habitat");
}
```

Runtime vs Compile-time

- Compiled language
- Compiler decides which method will be executed
- Compiler won't know which object actually is being used

```
public void makeSound(Animal a) {  
    System.out.println(a.getSound());  
}
```

Let's practice!

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Java Lists

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Java List

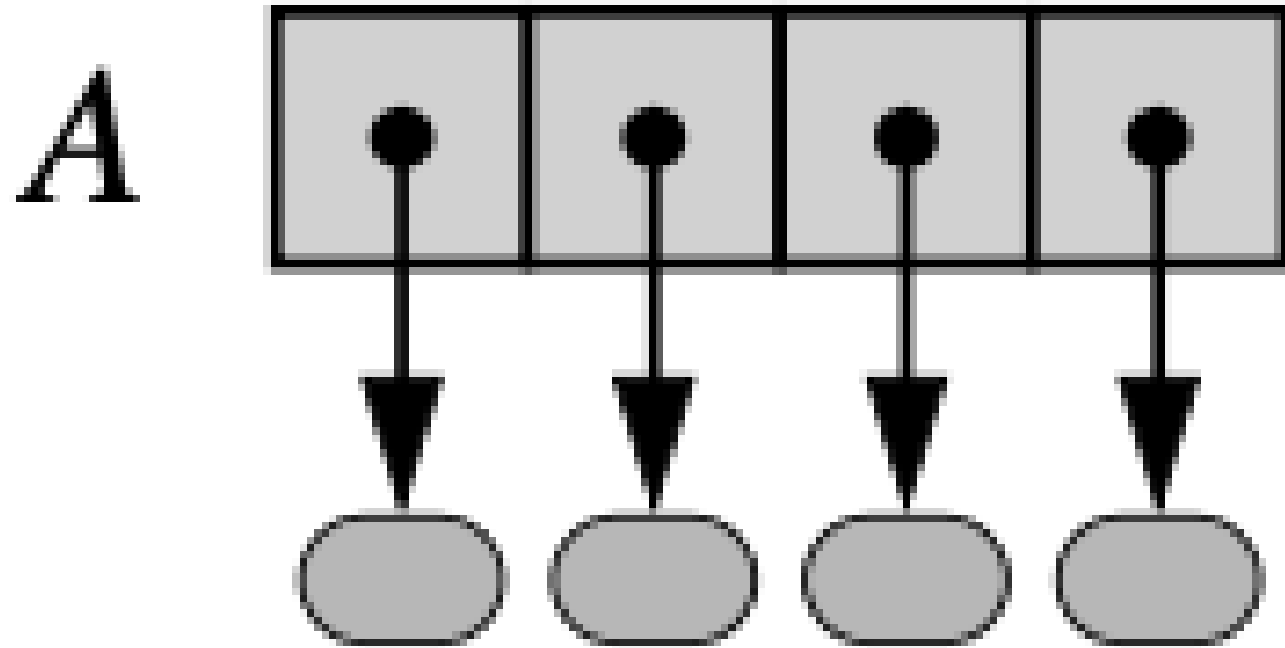
- `List` is an interface:

Method signatures:

- `int size()`
- `boolean isEmpty()`
- `boolean add(E e)`
- `boolean remove(Object o)`
- `E get(int index)`
- ...

`E` represents a generic data type

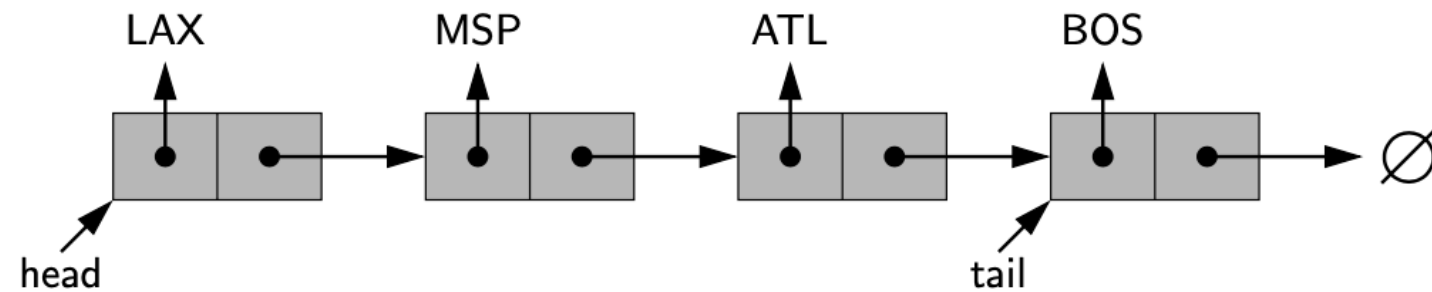
ArrayList



```
public E get(int index) {  
    Objects.checkIndex(index, size);  
    return elementData(index);  
}  
  
/unchecked/  
E elementData(int index) {  
    return (E) elementData[index];  
}
```

¹ Data Structures & Algorithms in Python, Goodrich

LinkedList



```
public E get(int index) {  
    checkElementIndex(index);  
    return node(index).item;  
}
```

```
    Returns the (non-null) Node at the specified element index.  
  
Node<E> node(int index) {  
    // assert isElementIndex(index);  
  
    if (index < (size >> 1)) {  
        Node<E> x = first;  
        for (int i = 0; i < index; i++)  
            x = x.next;  
        return x;  
    } else {  
        Node<E> x = last;  
        for (int i = size - 1; i > index; i--)  
            x = x.prev;  
        return x;  
    }  
}
```

¹ Data Structures & Algorithms in Python, Goodrich

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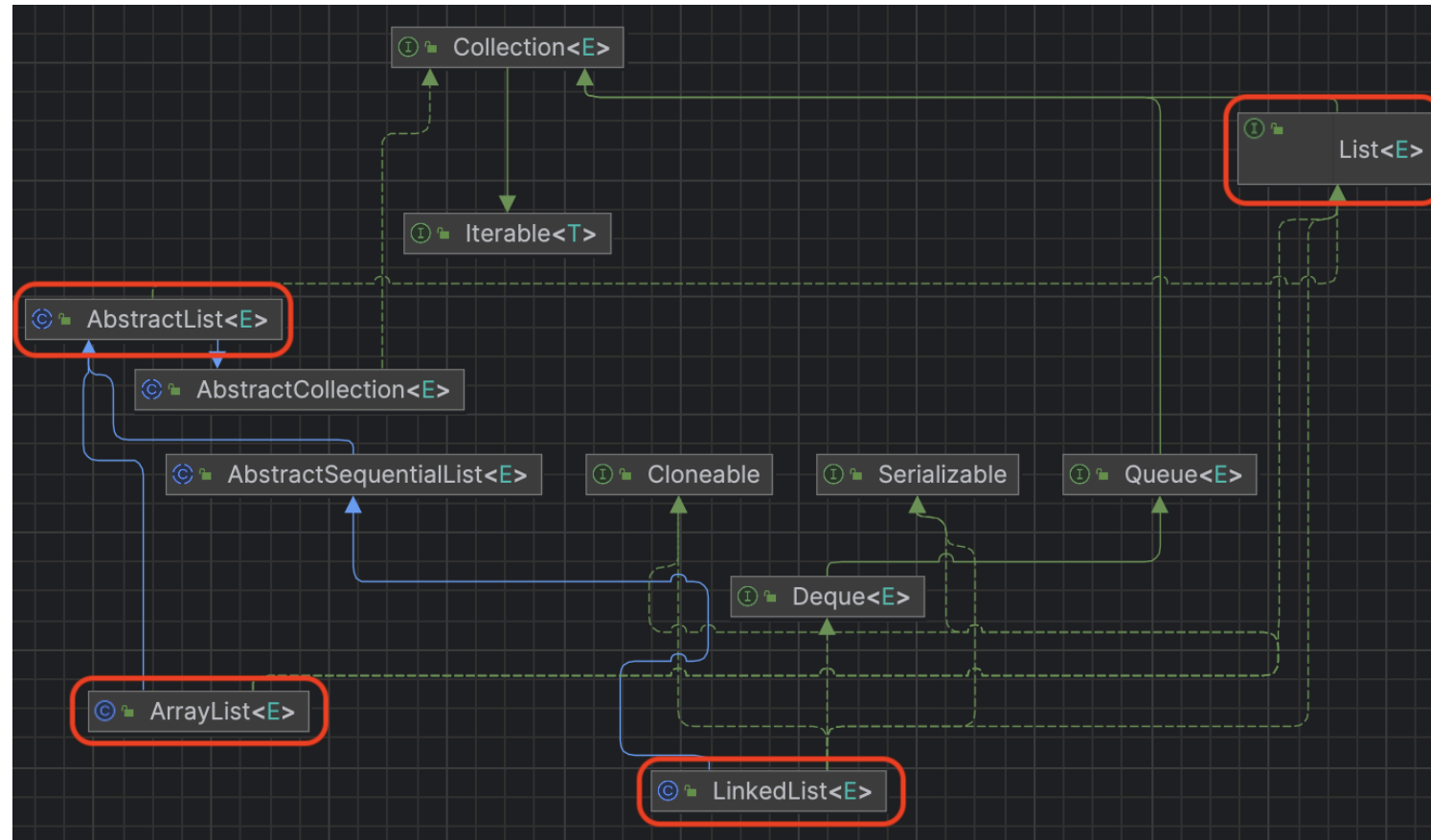
Abstract Lists

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Abstract Lists



- `equals()`
- `lastIndexOf()`
- Implemented at `AbstractList<E>`

Iterators



Method signatures for Iterator:

- `boolean hasNext()`
- `E next()`

Equals implementation

```
public boolean equals(Object o) {  
    if (o == this)  
        return true;  
    if (!(o instanceof List))  
        return false;  
  
    ListIterator<E> e1 = listIterator();  
    ListIterator<?> e2 = ((List<?>) o).listIterator();  
    while (e1.hasNext() && e2.hasNext()) {  
        E o1 = e1.next();  
        Object o2 = e2.next();  
        if (!(o1 == null ? o2 == null : o1.equals(o2)))  
            return false;  
    }  
    return !(e1.hasNext() || e2.hasNext());  
}
```

Let's practice!

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Wrap-up

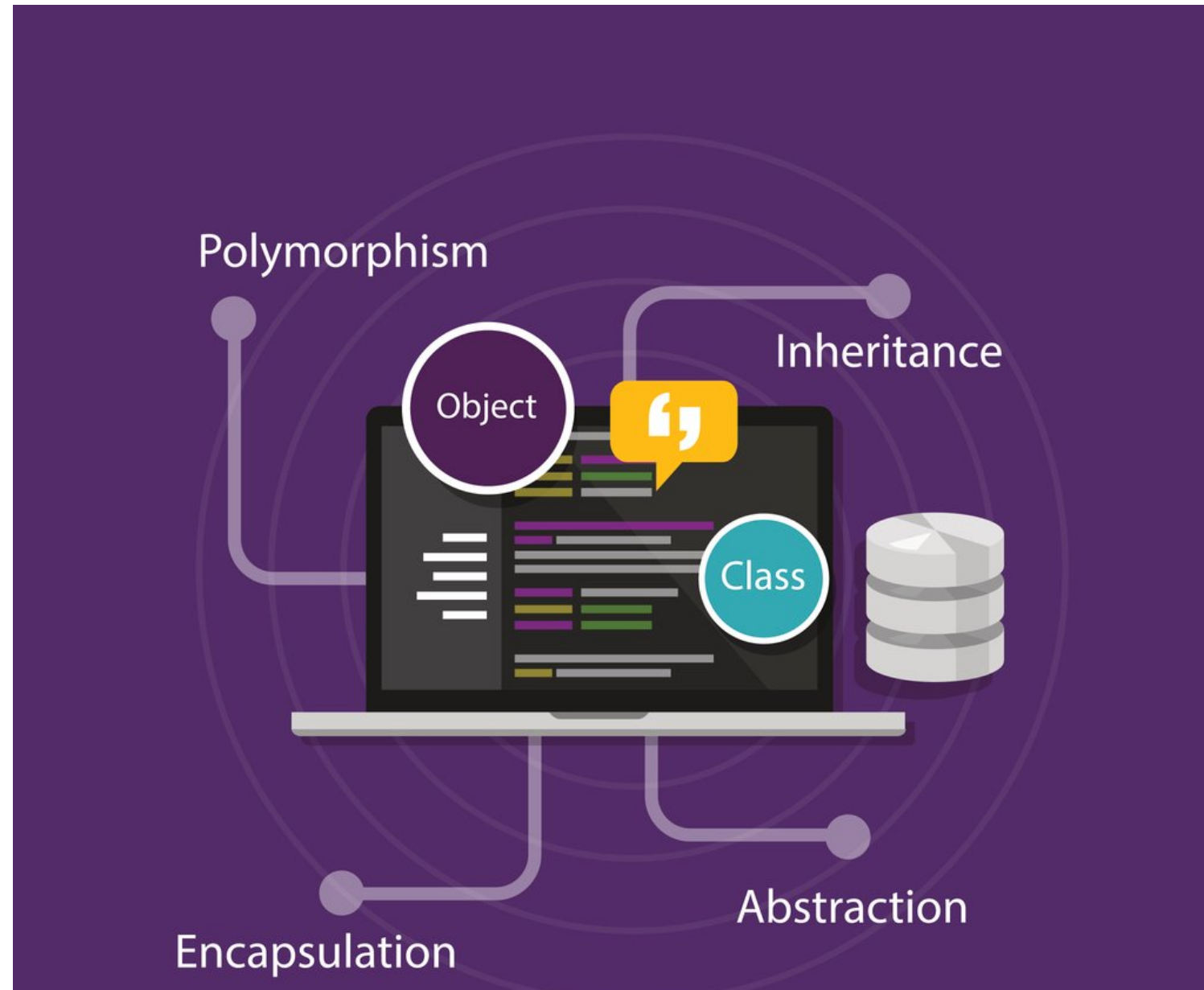
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Chapter 1 - Object-oriented programming (OOP) in Java



- What is OOP?
- Classes and objects
- Fields, constructors and methods

Chapter 2 - Implementing OOP in Java

- Access modifiers: `public` , `private` , `protected`
- Encapsulation
- Inheritance: `extends`
- Abstract classes: `abstract`
- Interfaces: `interface` , `implements`

Chapter 3 - Reviewing OOP implementations

- Polymorphism
- `List` , `ArrayList` , `LinkedList`

Next steps

- Internet Sources
 - **Baeldung**: website dedicated to Java topics and assistance
 - **Oracle's Java site**: the stewards of Java technology (Oracle) web site
 - **dev.java**: Java developer news, events, and other information
 - **Open JDK**: community dedicated to a free and open-source implementation of Java
- Books
 - *Head First Java* by Sierra et al
 - *Learn Java In One Day* by Jamie Chan
 - *Java for Beginners* by Brady Ellison

Congratulations!

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